

Fluoride Contamination in Groundwater from Parts of Anantpur District, Andhra Pradesh

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ABSTRACT

A total of 50 drinking water samples collected from Kadiri, Mudigubba, Nallamada revenue mandals of Anantapur district, Andhra Pradesh, India were analysed for fluoride contamination, besides water quality parameters such as pH, electrical conductivity, total dissolved solids, total alkalinity, total hardness, fluoride, bicarbonates, calcium, magnesium, nitrate, sulphate, phosphate, sodium and potassium. The results obtained were found to exceed the permissible limits. The concentration of fluoride in the water samples ranged between 0.5 mg/l and 5.5 mg/l and revealed that 90% of the water samples contain fluoride above the maximum permissible limit. The worst fluoride-affected villages are Ralla Anantapuram, Gunjepalli, Muktapuram, Nallamada, Toletivaripalli, Yerravankapalli, Kothapalli, Pathabathalapalli, Yerikireddipalli, Alampur, and Malakavemula. Geological formation is found to be a basic cause for the higher concentration of fluoride in most of the sampling points. Most of the people in these villages suffer from dental and skeletal fluorosis such as mottling of teeth, deformation of ligaments, bending of spinal column and ageing problem. The result of this study recommends that the groundwater should be used by the residents only after defluoridation.

Keywords: Fluoride, Groundwater, Dental and Skeletal Fluorosis, Anantapur District, Andhra Pradesh

INTRODUCTION

Groundwater though contributes only 0.6 % of the total water resources on earth, it accounts for nearly 80 % of the rural domestic water needs and 50 % of the urban water needs in the developing countries like India (Meenakshi and Maheshwari, 2006). Fluoride ion in drinking water is known for both beneficial and detrimental effects on health. It is essential for normal mineralization of bones and formation of dental enamel with presence in small quantity (Chouhan and Flora, 2010). However, when consumed in higher doses (>1.5 mg/l), it leads to dental fluorosis or mottled enamel and excessively high concentration (>3.0 mg/l) of fluoride may lead to skeletal fluorosis (Raju *et al.*, 2009).

Contamination of water resources with fluoride beyond acceptable limits is a health associated problem in many areas of south Asia and other regions of the world (Susheela, 1985; Karthikeyan *et al.*, 1996). The amount of fluoride present naturally in groundwater is governed by climate, pH, composition of the host rock and hydrogeology (Gupta *et al.*, 2006; Raju *et al.*, 2009). Some anthropogenic activities are also contributed to cause an increase in fluoride concentration in groundwater such as use of phosphatic fertilizers, pesticides, sewage and sludges, depletion of groundwater table etc (Ramanaiah *et al.*, 2006). Hence, to monitor the groundwater quality a lot of studies carried out throughout the world now a day (Subba Rao *et al.*, 2003; Nagarajan *et al.*, 2010; Gantam *et al.*, 2011; Kumar, 2011). Fluoride is considered as a major pollutant of groundwater on global scale. In the world, around 200 million people from 25 nations have great health risks, with high fluoride in drinking water. In the world, around 200 million people from 25 nations have great health risks, with high fluoride in the drinking water (Ayoob and Gupta, 2006). In the country (India), almost 60-65 million people drink fluoride contaminated groundwater and the number affected by fluorosis is estimated at 2.5 to 3 million in many states, especially, Andhra Pradesh, Bihar, Gujarat, Madhya Pradesh, Punjab, Rajasthan, Tamil Nadu and Uttar Pradesh (Athavale and Das, 1999; Susheela, 1999; Chakraborti *et al.*, 2000; Muralidharan *et al.*, 2002; Pillai and Stanley, 2002). The first case of endemic fluorosis in the country was reported as long as 1937 in the Prakasam District (erstwhile Guntur district), Andhra Pradesh (Shortt *et al.*, 1937). WHO (2004) has laid down a limit of 1.5 mg/l fluoride in drinking water. Groundwater is the major drinking water source in the villages of Anantapur district, Andhra Pradesh. The degree of fluorosis has been related to the concentration of fluoride in drinking water and period of consumption. The present paper reports on the prevalence of fluoride concentration in groundwaters of Kadiri, Mudigubba, Nallamada revenue mandals of Anantapur district, Andhra Pradesh, where the people rely on groundwater as potable water source. Some major

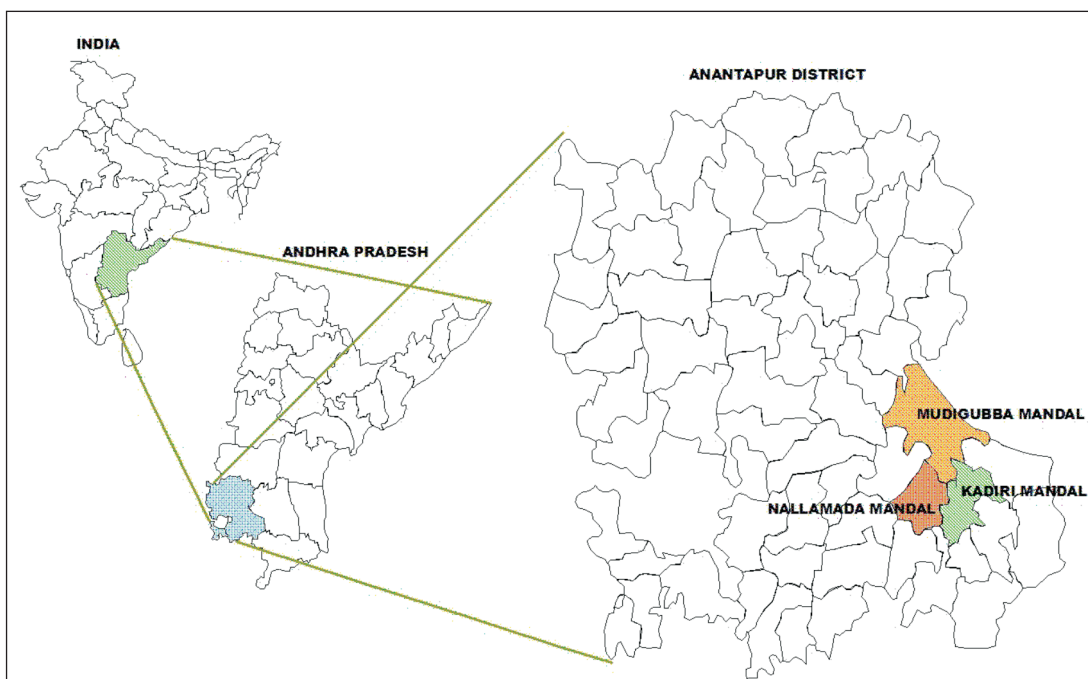


Fig.1. Location map of the study area.

inorganic constituents were also determined to ascertain the water quality.

STUDY AREA

The study area located in southeastern part of Anantapur district, Andhra Pradesh and lies between longitudes $77^{\circ} 30' - 78^{\circ} 15'$ East and latitudes $14^{\circ} 0' - 14^{\circ} 30'$ North and falls in the Survey of India toposheet Nos. 57 F/14, F/15, J/3, J/4 as shown in Fig. 1. The study area mainly exposes peninsular gneisses of Archean age consisting of pink granites, schists, composite gneisses of Dharwar age, intruded by a few pegmatite dykes and numerous dolerite dykes and the possible diamondiferous volcanic pipes. The geological map of the study area is shown in the Fig. 2. Geomorphic features observed from the study area are denudational hills, dissected pediments, pediplain, flood plain. The climate is semiarid with mean monthly temperatures of 15°C in January and 39°C in May. The normal rainfall of the district is 553 mm of which it secures the least rainfall as compared to Rayalaseema and other parts of Andhra Pradesh.

HYDROGEOLOGY

The district is underlain by granite gneisses and schists of Archean age. The northeastern part of the district is covered by consolidated sedimentary formation of Cuddapah Super Group belonging to upper Precambrian to lower Paleozoic age. River alluvium occurs along the major river courses and to some extent along minor stream courses. The Archean crystalline rocks include granites, gneisses and Dharwarian schists. The groundwater in these formations occurs in the

weathered and fractured zones under water table and semi-confined conditions respectively. These rock types do not possess primary porosity. Due to fractures and weathering, they have developed secondary porosity often giving rise to potential aquifers at depth. The degree of weathering in the Archean formation is less than 20 m. This weathered zone has been tapped extensively by the dug wells and dug-cum-bore wells, which invariably tap the fractures occurring below the weathered zone. Groundwater occurring in these formations is generally developed by dug wells, dug-cum-bore wells and bore wells. The depth of open wells range from 6.0 to 25.0 m below ground level and depth to water level vary from 1.5 to 23 m bgl. The yield of dug wells varies from 10 - 200 cm/day for a pumping period of 3 to 6 hrs. a day. The drainage map of the study area is shown in Fig. 3.

METHODOLOGY

Hundred groundwater samples were collected from 50 different locations of the study area which includes Kadiri, Mudigubba, Nallamada revenue mandals of Anantapur district, Andhra Pradesh, during October-November 2009. Sample location map of the study area is shown in Fig. 4. The samples were collected from bore wells which were extensively used for drinking and other domestic purposes. The samples were collected in pre-cleaned and sterilized polyethylene bottles of two litre capacity. The depth of the bore wells varied between 250 and 700 feet. The groundwater samples were analyzed using APHA (1995) procedure, and suggested precautions were taken to avoid contamination. The various parameters determined were pH, EC (electrical conductivity), total dissolved solids (TDS), total hardness

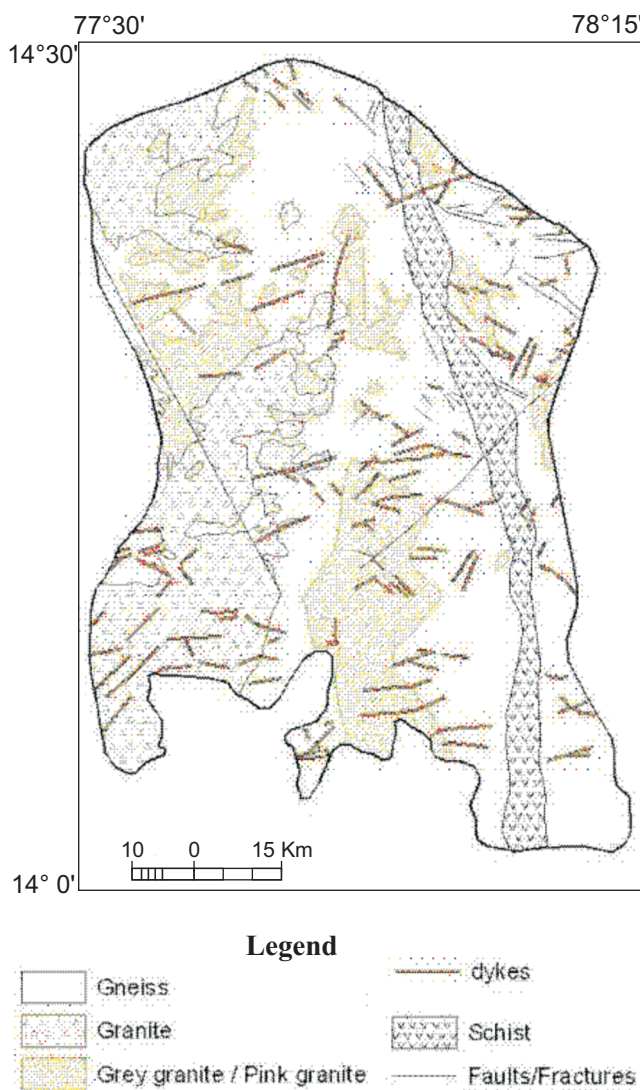


Fig. 2. Geological map of the study area.

(TH), calcium (Ca^{2+}), magnesium (Mg^{2+}), total alkalinity (TA), carbonate (CO_3^{2-}), bicarbonate (HCO_3^-), chloride (Cl^-), sulfate (SO_4^{2-}), Sodium (Na^+), potassium (K^+), Nitrate (NO_3^-) and Fluoride (F^-). pH and EC were determined by pH, conductivity meter, TDS by TDS meter, TH, Ca^{2+} , Mg^{2+} , CO_3^{2-} , HCO_3^- and Cl^- were estimated by titrimetry, where as Na^+ and K^+ by flame photometry (Systronics-128). F^- Was estimated by using ion selective electrode (Orion 4 star ion meter, Model: pH/ISE). All the experimental were carried out in triplicate and the results were found reproducible with in a $\pm 3\%$ error limit.

RESULTS AND DISCUSSION

The result of chemical analysis of fluoride concentration is presented in Table 1. Fluoride concentrations in the study area varied between 0.5 to 5.5 mg/l. Higher fluoride concentration (5.5 mg/l) was observed in the groundwater sample collected at

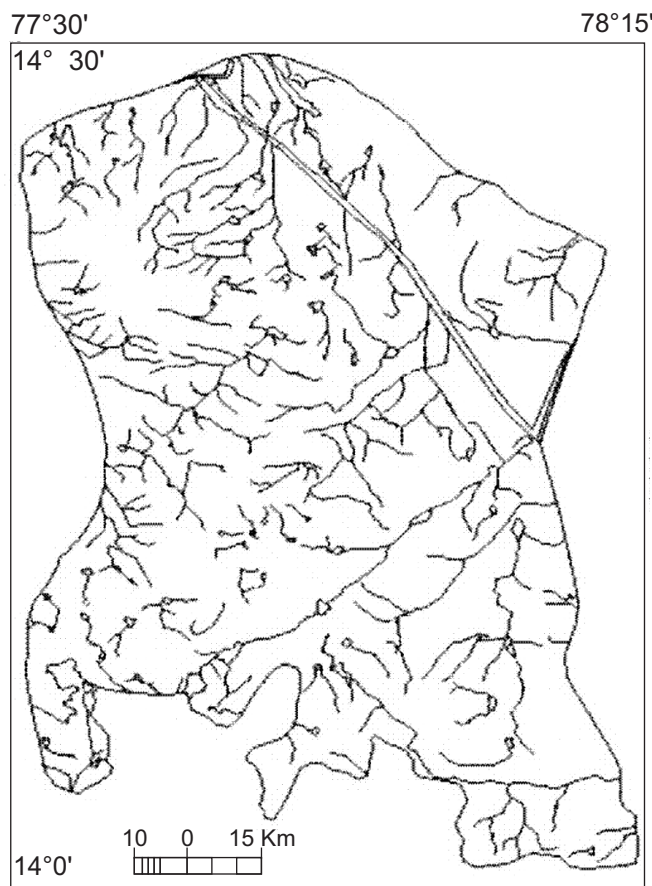


Fig. 3. Drainage map of the study area.

Nallamada village, while the lower concentration (0.5 mg/l) was observed at Reddipalli. About 90% of the samples of the study area are exceeding the permissible limits of fluoride 1.5 mg/l in drinking water WHO (2004). Fluoride concentration in natural waters depends on various factors such as temperature, pH, solubility of fluorine-bearing minerals, anion exchange capacity of aquifer materials (OH^- for F^-) and the nature of geologic formations drained by water and contact time of water with a particular formation. Minerals which have the greatest effect on the hydrogeochemistry of fluoride are fluorite, apatite, mica, amphiboles certain clays and villiamite. The aim of the present study is thus, to assess the possible sources of fluoride content in the groundwater for taking remedial measures.

Geochemistry of Fluoride

The details of the sampling sites are presented in Table 1. Fluoride concentration in natural waters depends on several factors such as temperature, pH, presence or absence of ion complexes or precipitation of ions and colloids, solubility of fluorine-bearing minerals, anion exchange capacity of aquifer materials (OH^- for F^-), the size and type of geological formations through which the water flows and the time water is in contact with a particular formation (Apambire *et al.*, 1997). Principally, controls are governed by climate, host rock composition

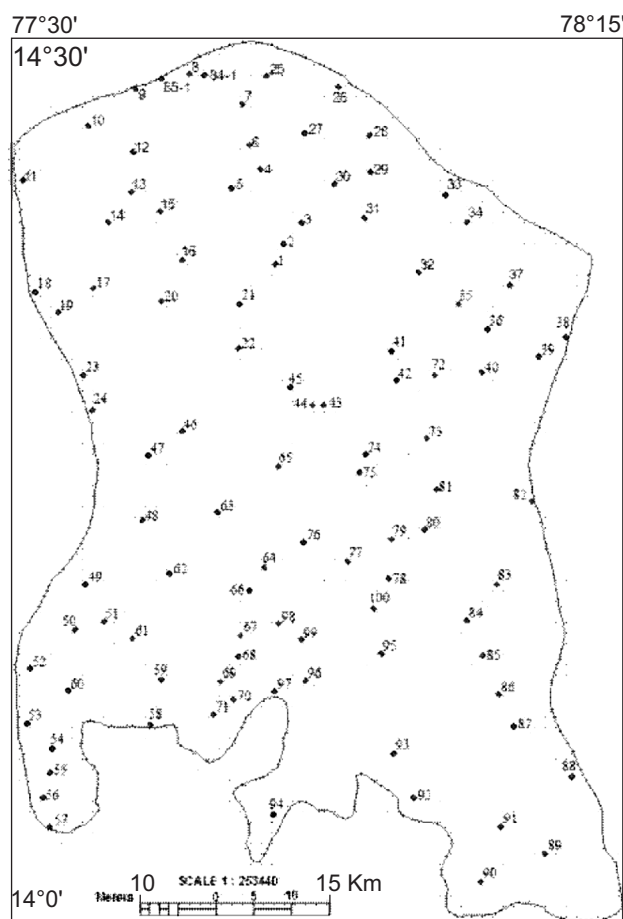


Fig. 4. Sample location map of the study area.

and hydrogeology. Areas of semiarid climate, crystalline rocks and alkaline soils are mainly affected (Sujatha, 2003). High fluoride concentrations groundwater suggest that favorable conditions exist for the dissolution of fluoride bearing minerals present in the granite and gneissic rocks in the study area. Fluoride bearing minerals occupy the joints, fractures, faults and vertical openings in the gneissic and granitic formations which are the oldest geological formations in Anantapur and have undergone maximum weathering (Rao, 2003). Granitic rocks are known to contain a relatively large proportion of fluorine minerals (Edmunds and Smedley, 2005). Fluorite, the main mineral that controls the geochemistry of fluoride in most environments is found in significant amounts in granite, granite gneisses and pegmatite (Deshmukh *et al.*, 1995; Rama Rao, 1982; Sujatha, 2003; UNICEF, 2008; Mamatha *et al.*, 2010). Analytical data revealed the presence of major anions and cations in the groundwater samples (Table 1). The data as in Table 1 shows significant variations in the water samples of chemical ingredients. The collected samples were not found to contain colour, odour and turbidity. The results were compared with WHO drinking water standards (Table 2). The statistical summary of the chemical constituents of the groundwater of the study area are shown in Table 3. The pH varies from 7 to 8.6, with a mean of 8 indicating an alkaline condition

which favours the solubility of fluoride-bearing minerals. In acidic medium (acidic pH), fluoride is adsorbed in clay; however, in alkaline medium it is desorbed, and thus alkaline pH is more favorable for fluoride dissolution activity (Saxena *et al.*, 2001). The electrical conductivity of the groundwater varies from 310 $\mu\text{S}/\text{cm}$ to 2580 $\mu\text{S}/\text{cm}$. Elevated concentration of electrical conductivity may possibly be credited to high salinity and high mineral content (Karthikeyan *et al.*, 2008). Total dissolved solids, a salinity indicator for the classification of groundwater, varies from 650 mg/l to 3280 mg/l in the study area. The bicarbonate content varies from 75 mg/l to 320 mg/l, and this high value indicates intense chemical weathering of the parent granite rock. Due to the common ion effect, the dissolution of fluorite is suppressed when the concentration of Ca is above the limit of fluorite solubility (Raju *et al.*, 2009). The calcium content in the groundwater of the study area varies from 70 mg/l to 360 mg/l and sodium values of groundwater are varying from 280 mg/l to 850 mg/l. The calcium concentrations were lower than the sodium concentration (Table 3), which indicates the higher fluoride concentration in the groundwater of the study area. A strong negative correlation between Ca^{2+} and F^- in the groundwaters that contain Ca^{2+} in excess of that required for the solubility of fluoride minerals has been observed by many researchers (Edmunds, 1992; Boyle, 1995). Generally high concentration of sodium will increase the solubility of fluoride bearing minerals in the waters. This may be the cause for the higher levels of fluoride in the groundwater of the study area. Under the prevailing semi-arid climatic conditions, during weathering of granites, gneissic rocks, fluorine is released from apatite and biotite to the circulating alkaline groundwater. In the study area fluoride contamination is mainly a natural process *i.e.* leaching of fluorine bearing minerals, since no man-made pollution has been noticed.

The concentrations of magnesium, chloride, sulfates were found to be 76, 211, 192 mg/l respectively, which were also considered to be above the permissible limit as prescribed by WHO. Excessive chlorides add bitter taste to water corrode steel and may cause cardio-vascular problems (Parameshwari and Kalpanadevi, 2006). High fluoride concentrations in parts of Mudigubba revenue mandal of Anantapur are mostly found in gneissic and granitic areas.

Enhanced nitrate concentration (20 mg/l to 110 mg/l), which was exceeding the permissible limits of 45 mg/l recommended by the WHO, was noticed in most of the samples. It is well known that the nitrogenous fertilizers are one of the important sources for groundwater nitrate for the past two decades. Many investigators have reported that the contribution of nitrate from the fertilizer to the groundwater can vary from as little as 3 mg/l to as much as 1800 mg/l (Malik and Banerji, 1981; Handa, 1983; Kolpin *et al.*, 1994). Further nitrogenous materials are rare in geological system (Uma, 1993). High NO_3^- concentrations may cause a potential fatal blood condition

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Table 1: Analytical data of the chemical analyses of groundwater in study area.

S.No	Names of Villages	pH	EC*	TDS	TH	Na ⁺	K ⁺	Ca ⁺²	Mg ⁺²	HCO ₃ ⁻	SO ₄ ⁻²	Cl ⁻	F ⁻	NO ₃ ⁻
1	Ralla Anantapuram	7.4	420	1040	678	720	60	170	70	170	240	140	4.28	58
2	Krishnapuram	7.6	620	1050	650	780	68	180	60	160	210	170	3.15	55
3	Nallacherlopalli	7.3	510	2470	655	800	70	200	50	140	200	190	2.49	60
4	Yerravankapalli	7.5	900	2450	346	850	72	70	58	150	140	170	2.65	68
5	Gunjepalli	8.3	850	2200	413	730	65	90	60	180	144	80	4.37	90
6	Jonnalakothapalli	8.2	700	1800	496	840	40	110	62	210	160	82	1.5	92
7	Muktapuram	8.3	350	1620	246	850	48	120	58	220	210	190	4.06	70
8	Krishnapuram	8.3	580	3020	505	780	50	130	50	230	220	95	3.93	30
9	Bapanakunta	8.3	310	1050	538	760	58	140	55	200	230	98	3.06	62
10	Reddipalli	8.4	340	3050	596	820	60	150	64	210	280	110	0.5	45
11	Vankarakunta	7.2	360	2020	662	810	72	160	72	180	300	120	1.89	48
12	Nallamada	7.0	600	2310	728	840	50	200	70	160	95	130	5.51	56
13	Pullagurapalli	8.0	680	1020	630	680	54	180	60	170	90	134	1.5	70
14	Toletivaripalli	8.3	650	1080	621	700	80	190	68	130	98	220	4.03	72
15	Donnekota	8.4	460	1000	637	820	85	170	76	120	100	210	1.51	56
16	Sankepal	8.5	920	880	494	810	40	150	80	90	120	190	0.54	48
17	Ramaswamy Tanda	8.6	900	850	478	520	42	80	82	80	150	94	0.65	98
18	Sanevaripalli	7.9	880	900	490	430	36	70	70	75	158	96	1.86	78
19	Rachavaripalli Tanda	8.2	620	920	596	410	50	90	60	95	140	110	1.57	65
20	Malakavemula	8.3	710	980	530	380	60	150	50	110	210	124	3.93	60
21	Nagurivandla Palli	8.0	760	1000	539	350	58	140	45	140	95	128	0.74	20
22	Alampur	8.6	630	1600	580	360	70	160	40	160	80	330	3.81	70
23	Vaddevari Palli	7.5	680	2000	639	310	72	180	42	150	210	410	3.47	100
24	K.Kuntlapalli	7.6	700	2600	540	280	78	200	30	130	220	210	1.49	46
25	Kondagattu Palli	8.2	720	1500	398	450	70	120	38	138	98	180	2.28	56
26	Chenchugaripalli Tanda	8.6	740	1100	480	540	55	140	40	140	150	170	1.5	90
27	Seelamvari Palli	8.0	650	1800	539	530	50	160	45	145	160	130	1.58	97
28	Ramireddi Palli	7.6	670	1600	621	600	54	180	50	160	180	170	2.81	88
29	Payagattu Palli	8.0	720	720	605	620	60	170	52	180	210	178	0.60	70
30	Yerikireddi Palli	8.2	340	780	704	580	62	180	60	200	200	180	3.98	66
31	Pandulakunta	8.3	410	1100	721	610	70	190	65	190	190	210	1.53	20
32	Kothannagari Palli	8.0	1020	950	537	600	68	110	70	220	220	220	2.49	68
33	Yerravanka Palli	7.4	1080	920	562	630	52	120	72	150	170	240	4.32	49
34	Podaralla Palli	7.6	780	650	528	570	40	100	75	140	160	160	2.83	56
35	Timmanayana Palem	7.2	800	680	478	580	30	70	80	160	180	72	2.72	70
36	Mallepalli	8.3	2580	3000	1240	320	160	260	160	280	290	378	0.55	100
37	Tappetavari Palli	8.0	860	700	548	710	38	90	82	200	190	240	1.53	77
38	Chiragarapalli	8.2	900	1010	594	700	54	110	84	210	220	260	1.56	56
39	Guttakindapalli Tanda	8.6	720	1020	628	650	52	125	86	230	210	270	1.53	72
40	Gollapalli	8.3	380	1220	648	840	50	130	90	210	140	290	1.55	54
41	Eddulavaripalli Tanda	8.0	400	1500	744	680	48	180	92	240	160	360	1.78	80
42	Choutakuntapalli	8.2	620	1620	835	700	45	200	98	250	220	310	1.72	80
43	Kothapalli	7.6	650	1700	910	720	50	220	110	260	240	280	4.57	63
44	R.Ramapuram	7.8	450	2000	926	680	70	210	120	264	250	210	3.77	33
45	Chandrayani Palli	7.6	520	2120	969	530	80	230	95	272	300	220	0.86	35
46	Pathabathala Palli	8.0	560	2140	1067	510	85	240	130	270	310	320	4.37	32
47	Uppalapadu	8.2	440	3100	1124	650	110	250	145	265	280	410	0.68	90
48	Malakavemula Cheruvu	8.5	2030	3240	1322	380	170	270	180	300	270	480	1.88	110
49	Nakkalaguttapalli	8.0	1250	3100	1333	370	150	320	130	280	260	450	1.51	90
50	Aakuthotapalli	8.2	1210	3280	1474	380	180	360	140	320	250	380	1.5	78

* EC in $\mu\text{S}/\text{cm}$, other parameters in mg/l .

Table 2: Comparison of the chemical parameters of groundwater of in Study Area with WHO standards for drinking purposes.

Parameters	WHO standards		Study area values	
	Max. Accept. limit	Max. Allow. limit	Minimum	Maximum
pH	6.5	8.5	7.0	8.6
EC	500	1500	310	2580
Ca ²⁺	75	200	70	360
Mg ²⁺	50	150	30	180
Na ⁺	50	200	280	850
K ⁺	-	200	30	180
HCO ₃ ⁻	30	150	75	320
Cl ⁻	200	600	72	480
SO ₄ ²⁻	200	400	80	310
F ⁻	1.0	1.5	0.5	5.5
NO ₃ ⁻	-	45	20	110

Table 3: Statistical Parameters of the different chemical constituents of groundwater in Study Area.

Parameters	Minimum	Maximum	Mean	Standard Dev
pH	7.0	8.6	8	0.40
Ec	310	2580	732.6	391.64
Ca ²⁺	70360	164.3	61.48	----
Mg ²⁺	30180	76.42	32.49	----
Na ⁺	280	850	615.2	168.24
K ⁺	30	180	67.82	32.36
SO ₄ ²⁻	20	110	192.16	60.80
Cl ⁻	72	480	211.98	102.86
HCO ₃	75	320	186.68	59.11
TDS	650	3280	1629.2	793.01
TH	246	1474	676.44	256.85
F ⁻	0.5	5.5	3.53	1.12
NO ₃ ⁻	20	110	65.94	21.02

known as methaemoglobinaemia, which especially affects infants.

The values of hardness vary from 246 mg/l to 1474 mg/l, around 80% of the samples were found to be very hard in nature. The high value of total hardness in supply water may cause corrosion of pipes, resulting in the pressure of certain heavy metals such as cadmium, copper, lead and zinc in drinking water. Total dissolved solids of the study area varies from 650 to 3280 mg/l. water with high total dissolved solids usually are lesser palatable and may bring in unfavorable physiological response, gastro-intestinal irritation. Elevated F⁻ in the groundwater was generally associated with low Ca²⁺ and high Na⁺ (Table 4). Na⁺ content is more than that of Ca²⁺, Mg²⁺ in water that may be due to CaCO₃ and MgCO₃ deposited and NaF is easily formed in the water solution. Thus it favours enrichment of fluoride in unconfined water. The Na⁺ acquired by groundwater from the

chemical weathering of rock forming minerals, is used as an index of the weathering of minerals alternate wet and dry conditions of the semiarid climate. This is responsible for leaching fluorine from the minerals present in the soils and rocks (Subba Rao *et al.*, 1993)

The approximate depth of the sources of the samples shows a positive correlation with the fluoride concentration (Fig. 5a) which indicates the source of fluoride to be fluorite or apatite minerals present in the Precambrian granite or granitic gneiss of the underground basement.

The correlation of some selected ions and other parameters with fluoride are shown in Fig. 5b to f. Positive correlation between fluoride content and pH in the groundwater suggest that pH is important in determining fluoride in groundwater in agreement with earlier observations (Gupta *et al.*, 1986) as found previously by others (Handa, 1975). The alkali metal ions,

Table 4: Correlation coefficients of chemical parameters of the study area.

	pH	EC	TDS	TH	Na ⁺	K ⁺	Ca ⁺²	Mg ⁺²	HCO ₃ ⁻	SO ₄ ²⁻	Cl ⁻	F ⁻
EC	-0.218	1										
TDS	-0.061	0.456	1									
TH	-0.029	0.229	0.605	1								
Na ⁺	0.076	0.463	0.53	0.342	1							
K ⁺	-0.112	0.338	0.344	0.208	0.186	1						
Ca ⁺²	-0.045	0.101	0.119	0.078	-0.053	-0.01	1					
Mg ⁺²	0.066	-0.006	0.01	0.436	0.196	-0.014	-0.09	1				
HCO ₃ ⁻	-0.068	0.083	0.034	0.029	-0.041	0.103	0.031	0.042	1			
SO ₄ ²⁻	-0.02	0.419	0.36	0.217	0.591	0.06	0.02	0.203	-0.007	1		
Cl ⁻	-0.443	0.17	0.061	-0.044	0.02	0.07	0.216	-0.038	-0.003	-0.079	1	
F ⁻	0.104	0.078	0.211	0.243	0.395	0.037	0.022	0.237	0.160	0.202	0.055	1
NO ₃ ⁻	-0.029	0.345	0.177	0.094	0.107	0.018	0.151	-0.128	-0.058	0.105	0.092	0.162

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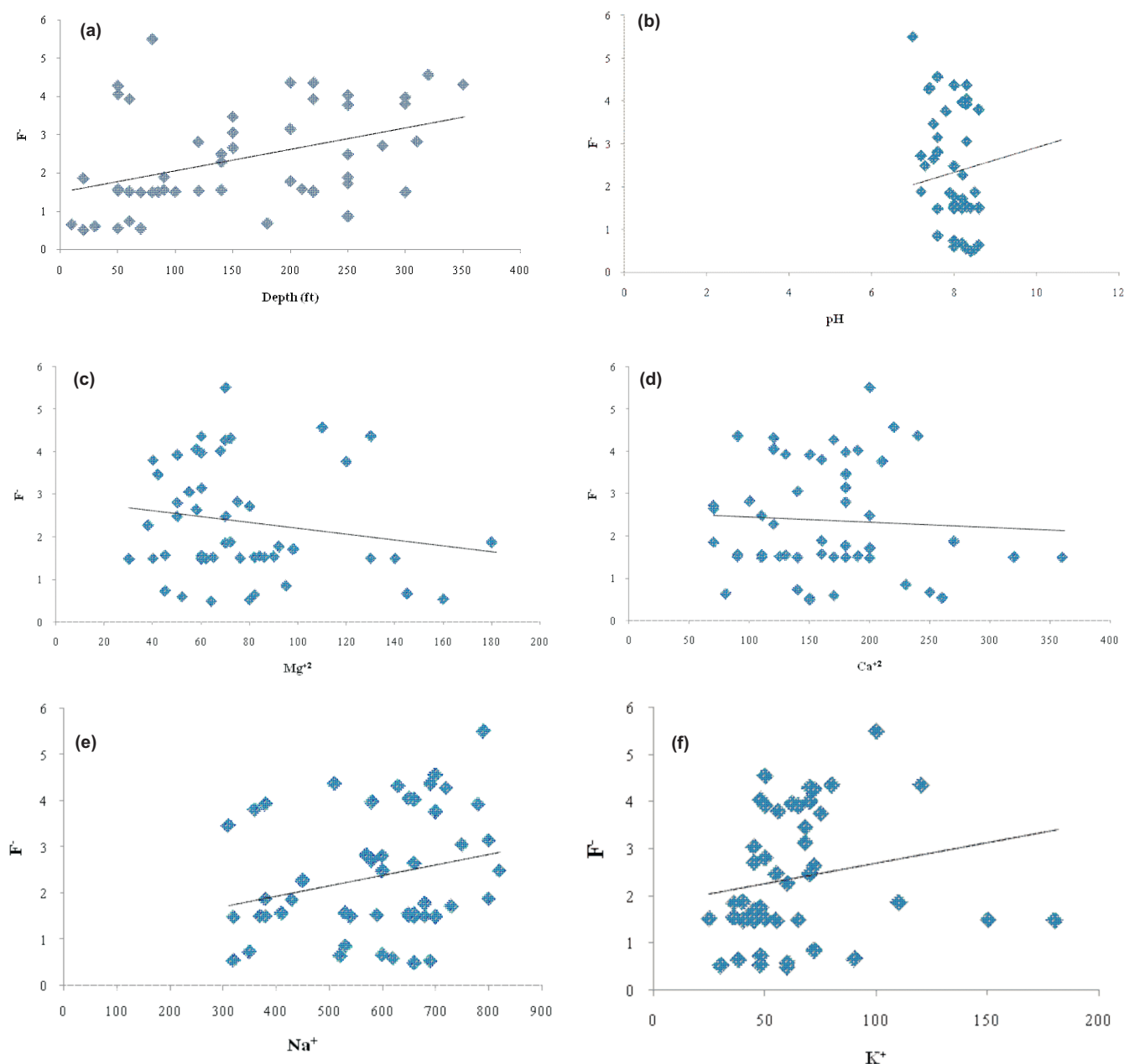


Fig.5. Correlation of depth different ions and other parameters with F^- concentration in selected groundwater samples: (a) pH vs F^- (b) Mg^{+2} vs F^- (c) Ca^{+2} vs F^- (d) Na^+ vs F^- (e) K^+ vs F^- .

viz. Na^+ , K^+ showed positive correlation with fluoride content with correlation coefficient of 0.606, 0.412 respectively. The alkaline earth metal ions *viz.* Mg^{+2} and Ca^{+2} showed negative correlation with fluoride content. The correlation coefficients were -0.391, -0.548 respectively Fig. 5a. Correlations with other parameters however are not statistically significant.

Impact of Fluoride on Human Health

In general, fluoride content in water between 1.5 and 2.0 mg/l may lead to dental mottling, which is characterized initially by opaque white patches on the teeth and in advanced stages leads to dental fluorosis (Pillai and

Stanley, 2002) followed by pitting of teeth surfaces. High manifestations of dental fluorosis are mostly found in children up to the age of 12 years.

Skeletal fluorosis may occur when fluoride concentrations in drinking water exceed 4-8 mg/l. The symptoms include mild rheumatic/arthritis pain in joints and muscles to severe pain in the cervical spine region along with stiffness and rigidity of the joints (Teotia and Teotia, 1988) while excess F^- may include hypocalcaemia (Sherlin and Verma, 2000; Pius, 1999; Ekambaram and Vanaja, 2001). Crippling skeletal fluorosis can occur when the water supply contains more than 10 mg/l of fluoride (WHO, 1984; Boyle and Chignon, 1995).

CONCLUSIONS

The Anantapur district of Andhra Pradesh state, India, is a typical region showing endemic fluorosis caused by drinking water. Groundwater (well, hand pump and tube well) is the main source of drinking water for village residents. Geological formation is found to be a basic cause for the higher concentration of fluoride in most of the sampling points. The present study concludes that alkaline environment is the dominant controlling factor for leaching of fluoride from the source material in the groundwater. Most of the chemical concentrations collected from the study area do not comply with the water quality standards. Several reports on dental and skeletal manifestations of fluorosis are also reported in the study area, which shows that the population of the study area is at higher risk due to excessive fluoride

intake. The worst fluoride affected villages are Ralla Anantapuram, Gunjepalli, Muktapuram, Nallamada, Toletivaripalli, Yerravankapalli, Kothapalli, Pathabathalapalli, Yerikire-ddipalli, Alampur, Malakavemula. Most people in these villages suffer from dental and skeletal fluorosis such as mottling of teeth, deformation of ligaments, bending of spinal column and ageing problem. An urgent need is to educate the people on the causes of fluorosis, and providing fluoride free drinking water in the study area.

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REFERENCES

- Anandha Parameshwari, N. and Kalpanadevi, K. (2006). Groundwater analysis of Rajavoor, Udumalpet- A case study. *In: Proc. of national conference on environmental degradation and pollution control*, Coimbatore, India.
- Apambire, W.B., Boyle, D.R. and Michel, F.A. (1997). Geochemistry, genesis, and health implications of fluoriferous groundwater in the upper regions of Ghana. *Env. Geol.*, v.33(1), pp.13-24.
- Athavale, R.N. and Das, R.K. (1999). Beware! Fluorosis is zeroing in on you. *Down to Earth*, v.8, pp.24-25.
- Ayoob, S. and Gupta, A.K. (2006). Fluoride in drinking waters: a review on the status and stress effects. *Critical Rev. Env. Sci. Tech.*, v.36, pp.433-487.
- BIS (2003). Indian standard specifications for drinking water. New Delhi, India: IS:10500, Bureau of Indian Standards, 10p.
- Boyle, D.R. and Chagnon, M. (1995). An incident of skeletal fluorosis associated with groundwaters of the maritime carboniferous basin, Gaspé region, Quebec, Canada. *Env. Geochem. Health*, v.17, pp.5-12.
- Chakraborti, D., Chanda, C.R., Samanta, G., Chowdary, U.K., Mukherjee, S.C. and Pal, A.B. (2000). Fluorosis in Assam, India. *Curr. Sci.*, v.78, pp.1421-1423.
- Chouhan, S. and Flora, S.J.S. (2010). Arsenic and Fluoride: Two Major Groundwater Pollutants. *Indian Jour. Exp. Bio.*, v.48, pp.66-678.
- Deshmukh, A.N., Wadaskar, P.M. and Malpe, D.B. (1995). Fluoride in environment: a review. *Gond. Geol. Mag.*, v.9, pp.1-20.
- Edmunds, M. and Smedley, P. (2005). Essentials of medical geology-Impacts of the natural environment on public health, *In: O. Selinus, B. Alloway, J.A. Centeno, R.B. Finkleman, R. Fuge and U. Lindh (Eds.), Fluoride in natural waters*, Academic Press, Amsterdam.
- Edmunds, W.M. (1992). Hydro-geochemistry of groundwaters in the Derbyshire dome with special reference to trace constituents. Institute of Geological Science, Great Britain, Report 71/7.
- Ekambaram, P. and Vanaja, P. (2001). Calcium preventing locomotor behavioral and dental toxicities of fluoride by decreasing serum fluoride levels in rats. *Env. Toxic.*
- Pharm., v.9, pp.141-146.
- Gupta, S., Banerjee, S., Saha, R., Datta, J.K. and Mondal, N. (2006). Fluoride geochemistry of groundwater in Nalhati-1 block of the Birbhum district, West Bengal, India. *Fluoride*, v.39(4), pp.318-320.
- Gautam, R., Bhardwaj, N. and Saini, Y. (2011). Study of fluoride content in groundwater of Nawa tehsil in Nagaur, Rajasthan. *Jour. Env. Bio.*, v.32(1), pp.85-89.
- Handa, B.K. (1983). Effect of fertilizer on groundwater quality in India. *In: symposium on groundwater development -A perspective for the year 2000 A.D.*, Roorkee, India.
- Karthikeyan, G., Anitha, P. and Apparao, B.V. (1996). Contribution of fluoride in water and food to the prevalence of fluorosis in areas of Tamil Nadu in south India. *Fluoride*, v.29(3), pp.151-155.
- Kolpin, D.W., Burkart, M.R. and Thurman, E.M. (1994). Herbicides and nitrate in near surface aquifers in the mid continental United States. *U.S. Geological Survey Water Supply Papers*, 2413, pp.1-34.
- Kumar, N. (2011). Variation of fluoride and correlation with alkalinity in groundwater of shallow and deep aquifers. *Int. Jour. Env. Sci.*, v.1(5), pp.884-890.
- Malik, S. and Banerjee, S. (1981). Nitrate pollution of groundwater as a result of agriculture development in Indo-Ganga Plain, India. *In: Proceedings of international symposium on quality of groundwater*, Netherlands.
- Mamatha, P. and Sudhakar Rao, M. (2010). Geochemistry of fluoride rich groundwater in Kolar and Tumkur Districts of Karnataka. *Environ. Earth Sci.*, v.61, pp.131-142.
- Meenakshi, R.C. and Maheshwari (2006). Fluoride in drinking water and its removal. *Jour. Hazardous Mat.*, v.B137, pp.456-463.
- Muralidharan, D., Nair, A.P. and Satyanarayana, U. (2002). Fluoride in shallow aquifers in Rajgarh tehsil of Churu district, Rajasthan - An arid environment. *Curr. Sci.*, v.83, pp.699-702.
- Nagarajan, R., Rajmohan, N., Mahendran, U. and Senthil kumar, S. (2010). Evaluation of groundwater quality and its suitability for drinking and agricultural use in Thanjavur city, Tamil Nadu, India. *Environ. Monit. Assess.*, v.171, pp.289-308.
- Pillai, K.S. and Stanley, V.A. (2002). Implications of fluoride -

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- an endless uncertainty. *Jour. Env. Bio.*, v.23, pp.81-87.
- Pius, A., Appa Rao, B.V. and Karthtkeyan, G. (1999). Role of calcium in the amelioration of fluorosis a case study; *Indian Jour. Env. Health*, v.41, pp.121-125.
- Raju, N.J., Dey, S. and Das, K. (2009). Fluoride contamination in groundwaters of Sonbhadra district, Uttar Pradesh, India. *Curr. Sci.*, v.96(7), pp.979-985.
- Ramanaiah, S.V., Venkatamohan, S., Rajkumar, B. and Sarma, P.N. (2006). Monitoring of fluoride concentration in groundwater of Prakasham district in India: Correlation with physico-chemical parameters. *Jour. Env. Sci. Eng.* v.48, pp.129-134.
- Rama Rao, N.V. (1982). Geochemical factors influencing the distribution of fluoride in rocks, soils and water sources of Nalgonda district, A.P. Ph.D. Thesis, Osmania University, Hyderabad.
- Rao, N.S. (2003). Groundwater quality: focus on fluoride concentration in rural parts of Guntur district, Andhra Pradesh, India. *Hydrol. Sci. Jour.*, v.48, 35p.
- Saxena, V.K. and Ahmed, S. (2001). Dissolution of fluoride in groundwater: a water-rock interaction study. *Environ. Geol.*, v.40, pp.1084-1087.
- Sherlin, D.M.G. and Verma, R.J. (2000). Amelioration of fluoride induced hypocalcaemia by vitamins. *Human Exp. Toxic.*, v.19, pp.632-634.
- Shortt, H.E., McRobert, G.R., Barnard, T.W. and Mannadinayer, A.S. (1937). Endemic fluorosis in Madras Presidency, India. *Indian Jour. Med. Res.*, v.25, pp.553-561.
- Subba Rao, N. and Rao, A.T (2003). Fluoride in groundwaters in a developing area of Guntur district, Andhra Pradesh, India. *Jour. Appl. Geochem.*, v.5, pp.94-100
- Subba Rao, N. and Devadas, D. (2003). Fluoride incidence in groundwater in an area of peninsular India. *Environ. Geol.*, v.45, pp.243-253.
- Sujatha, D. (2003). Fluoride levels in the groundwater of the south-eastern part of Ranga Reddy district, Andhra Pradesh, India. *Environ. Geol.*, v.44, pp.587-591.
- Susheela, A.K. (1985). Epidemiology and control of Fluorosis in India. *Fluoride*, v.18(2), pp.120-121.
- Susheela, A.K. (1999). Fluorosis management programme in India. *Curr. Sci.*, v.77, pp.1250-1256.
- Teotia, S.P.S. and Teotia, M. (1988). Endemic skeletal fluorosis clinical and radiological variants. *Fluoride*, v.21, pp.39-44.
- Uma, K.O. (1993). Nitrates in shallow (regolith) aquifers around Sokoto town, Nigevia. *Environ. Geol.*, v.21, pp.70-76.
- UNICEF (2008). UNICEF handbook on water quality. United Nations Children's Fund, New York.
- WHO (1984). The guidelines for drinking water quality recommendations. World Health Organization, Geneva.
- WHO (2004). Guidelines for drinking water quality. 3rd Edn., World Health Organization, Geneva.